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Project Guide

AGENDA

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Problem Statement

To design and develop an agentic AI agent for GATE aspirants that dynamically recommends targeted practice questions, utilizing real-time performance analytics to adaptively challenge students and bridge specific conceptual gaps.

Need of the System

- Current GATE preparation platforms mostly provide static question banks that do not adapt to individual student performance.
- Students often waste time practicing random questions instead of focusing on their weak concepts.
- Traditional systems lack real-time performance analysis to identify specific conceptual gaps.
- Many aspirants struggle to track their progress and understand which topics require more practice.
- There is limited availability of personalized learning support for each student's pace and ability.
- An AI-based system can analyze performance data and recommend targeted practice questions.
- Adaptive difficulty can continuously challenge students while reinforcing weaker areas.
- Such a system can improve learning efficiency, concept clarity, and exam readiness for JEE Main aspirants.

Abstract

Preparing for the JEE Main exam requires strong conceptual understanding and continuous practice. However, most existing platforms provide static question banks that do not adapt to individual student performance, making preparation less efficient.

This project proposes the development of an **autonomous AI agent** that recommends targeted practice questions for JEE Main aspirants. The system analyzes student performance in real time to identify conceptual gaps and suggest questions with appropriate difficulty levels.

By providing adaptive and personalized practice, the system helps students focus on weak areas, improve problem-solving skills, and enhance overall exam preparation.



Literature Survey - Related to Existing Systems

Existing Solutions

- **National Test Abhyas** – An adaptive learning platform developed by the National Testing Agency that provides daily mock tests and performance analytics for JEE and NEET aspirants. It helps students analyze their scores and improve through regular practice tests.
- **eTutor Adaptive Practice** – A platform offering adaptive practice where the difficulty of questions changes based on the student's answers, helping reinforce weak topics and progressively improve learning.
- **QANDA** – An app that allows students to upload a photo of a question and receive instant solutions, mainly used for quick doubt solving during JEE preparation.
- **Khanmigo (by Khan Academy)** – A tutoring assistant that guides students through problem-solving and provides personalized learning support instead of simply giving direct answers.

Literature Survey (in form of table)

Paper Name	Journal/Conference Name and Year/publishing house	Methodology	GAPS
Agentic AI-based question generation system using Retrieval-Augmented Generation	IEEE	Data Input Collection Data Processing and Knowledge Extraction Vector Embedding and Database Creation Query Processing Question Generation	Limited syllabus or curriculum alignment Lack of structured exam paper generation Limited multi-step intelligent workflow Limited personalization
Adaptive AI agent system for personalized learning frameworks and practical applications in education.	IEEE	Multi-modal AI Agent framework Dynamic content delivery Adaptive Assessment System Engaging Learning Paths Automated Design of Agentic Systems	Lack of personalization Weak integration of educational theories
Agentic AI in Inclusive Learning: A Framework for Autonomous Personalization across Diverse Learner Populations	Google Scholar	AI education review SDT-UDL foundation Adaptive vs Agentic comparison 10-pillar AI framework Multi-layer architecture design Implementation challenge analysis	Existing AI learning systems lack personalization. Lack of real-world validation of agentic AI frameworks in education.





Gaps Identified

Gaps in Existing Systems

- **eTutor Adaptive Practice:** Adjusts difficulty levels but offers limited deep conceptual gap analysis.
- **QANDA:** Primarily a doubt-solving tool without long-term performance tracking or adaptive practice.
- **National Test Abhyas:** Focuses mainly on mock tests and lacks continuous real-time adaptive question recommendation.
- **Khanmigo (Khan Academy):** Designed for general learning and not specifically optimized for JEE exam patterns.

Proposed solution

- Develop an **autonomous AI-based learning agent** that assists JEE Main aspirants in practicing questions more efficiently.
- The system will **analyze student performance in real time** based on accuracy, response time, and topic-wise results.
- Using this analysis, the AI agent will **identify conceptual gaps and weak areas** in subjects such as Physics, Chemistry, and Mathematics.
- The agent will **dynamically recommend targeted practice questions** using PYQs and syllabus.
- The system will implement **adaptive difficulty**, gradually increasing or decreasing question complexity based on student performance.
- A **performance analytics dashboard** will help students track progress, strengths, and improvement areas.
- This solution aims to provide **personalized, efficient, and data-driven preparation** for JEE Main aspirants.



Flow of System

PERSONALISED LEARNING JOURNEY: SYSTEM FLOWCHART



Data sources

- Previous Year Questions
- Syllabus
- <https://gateoverflow.in/422782/gate-cse-2024-set-1-ga-question-5>

Conclusion

The proposed system aims to enhance JEE Main preparation by providing a personalized and adaptive learning experience. By using an autonomous AI agent to analyze student performance and recommend targeted practice questions, the system helps identify conceptual gaps and focus on weaker areas. This approach makes practice more efficient, improves problem-solving skills, and supports students in achieving better exam readiness.



THANK YOU !!

